King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 233

Original exam date: 25 July 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $h(x) = \frac{1}{9}(3x+1)^3$, then $h''(1) =$

(a) 24

(b) 28

(c) 26

(d) 22

(e) 20

2. If $y = Ax + B$ is the equation of the normal line to the circle $x^2 + y^2 = 25$ at the point $(4, 3)$, then $A + B =$

(a) $\frac{-4}{3}$

(b) $\frac{1}{5}$

(c) $\frac{-3}{4}$

(d) $\frac{3}{4}$

(e) $\frac{4}{3}$

3. The value of c that satisfies Rolle's Theorem when applied to $f(x) = (x-2)(x+3)^2$ on $[-3, 2]$ is equal to

(a) 2

(b) $\frac{1}{3}$

(c) -3

(d) $-\frac{1}{3}$

(e) $\frac{1}{2}$

4. If $g(x) = \left(\dfrac{3x^2-2}{2x+3}\right)^{-2}$, then $g'(0) =$

(a) 0

(b) 3

(c) 4

(d) 1

(e) 2

5. The slope of the graph of the relation $x + y - 1 = \ln(x^2 + y^2)$ at the point $(1, 0)$ is

(a) -1

(b) 1

(c) -2

(d) 0

(e) 2

6. If $y = \arctan(3x) + \arcsin(\sqrt{x})$, then $\left.\dfrac{dy}{dx}\right|_{x=\frac{1}{2}} =$

(a) $\dfrac{21}{13}$

(b) $\dfrac{27}{13}$

(c) $\dfrac{29}{13}$

(d) $\dfrac{23}{13}$

(e) $\dfrac{25}{13}$

7. The value of c that satisfies the Mean Value Theorem when applied to $f(x) = x - \cos x$ on $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ is equal to

(a) 0

(b) $-\dfrac{\pi}{4}$

(c) $\dfrac{\pi}{6}$

(d) $-\dfrac{\pi}{3}$

(e) $\dfrac{\pi}{3}$

8. A point moves along the curve $y = \sqrt{x}$ in such a way that the y-component of the position of the point is increasing at a rate of 2 units per second. At what rate is the x-component changing when $x = 4$?

(a) increasing at a rate of $\frac{1}{2}$ unit per second

(b) decreasing at a rate of 8 units per second

(c) decreasing at a rate of $\frac{1}{2}$ unit per second

(d) increasing at a rate of 1 unit per second

(e) increasing at a rate of 8 units per second

9. Newton's Method is used to approximate a zero of the function $f(x) = x - e^{-x}$. If we choose $x_1 = 0$, then $x_2 =$

(a) -1

(b) $\frac{1}{2}$

(c) $-\frac{1}{2}$

(d) 1

(e) 2

10. If $y = (\ln x)^{\ln x},\ x > 1$, then $y'(e) =$

(a) e

(b) 1

(c) e^2

(d) $\frac{1}{e}$

(e) $\frac{1}{e^2}$

11. An airplane is flying at an altitude of 5 miles and passes directly over a radar antenna. When the distance between the plane and the radar is 10 miles, the radar detects that the distance between the radar and the plane is changing at a rate of 240 miles per hour. What is the speed of the plane?

(a) $320\sqrt{3}$

(b) 160

(c) 480

(d) $160\sqrt{3}$

(e) 320

12. If M and m are the absolute maximum and absolute minimum respectively of the function $f(x) = 3x^{2/3} - 2x$ on $[-1, 1]$, then $M + m =$

(a) 2

(b) 5

(c) 3

(d) 4

(e) 7

13. If $x = c$ is a critical number of the function $f(x) = \sin^2 x + \cos x$ on $(0, 2\pi)$, then the sum of all possible values of c is

(a) π

(b) 3π

(c) 2π

(d) $\dfrac{2\pi}{3}$

(e) $\dfrac{5\pi}{3}$

14. If $f(x) = \tan x,\ -\dfrac{\pi}{4} \leq x \leq \dfrac{\pi}{4}$, then $\left(f^{-1}\right)'\left(\dfrac{\sqrt{3}}{3}\right) =$

(a) $\dfrac{1}{2}$

(b) $\dfrac{5}{4}$

(c) 1

(d) $\dfrac{1}{4}$

(e) $\dfrac{3}{4}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$h' = (3x+1)^2$, $h'' = 6(3x+1)$, so $h''(1) = 24$.
2	**(d)**	Tangent slope $-\frac{4}{3}$, normal slope $\frac{3}{4}$, line $y = \frac{3}{4}x$: $A + B = \frac{3}{4}$.
3	**(b)**	$f' = (x+3)(3x-1) = 0$ with $c \in (-3, 2)$ gives $c = \frac{1}{3}$.
4	**(b)**	$g' = -2u^{-3}u'$ with $u(0) = -\frac{2}{3}$, $u'(0) = \frac{4}{9}$: $g'(0) = 3$.
5	**(b)**	$1 + y' = \frac{2x+2yy'}{x^2+y^2}$, at $(1, 0)$: $1 + y' = 2$, so $y' = 1$.
6	**(e)**	$y' = \frac{3}{1+9x^2} + \frac{1}{2\sqrt{x(1-x)}}$, at $x = \frac{1}{2}$: $\frac{12}{13} + 1 = \frac{25}{13}$.
7	**(a)**	Average rate $= 1$, $f'(c) = 1 + \sin c = 1$, so $c = 0$.
8	**(e)**	$\frac{dy}{dt} = \frac{1}{2\sqrt{x}}\frac{dx}{dt}$: $2 = \frac{1}{4}\frac{dx}{dt}$, so $\frac{dx}{dt} = 8$ (increasing).
9	**(b)**	$x_2 = 0 - \frac{f(0)}{f'(0)} = 0 - \frac{-1}{2} = \frac{1}{2}$.
10	**(d)**	$\frac{y'}{y} = \frac{\ln(\ln x)+1}{x}$; at $x = e$, $y = 1$: $y'(e) = \frac{1}{e}$.
11	**(d)**	$x\,x' = z\,z'$ with $x = 5\sqrt{3}$, $z = 10$, $z' = 240$: speed $= 160\sqrt{3}$.
12	**(b)**	Values $f(-1) = 5$, $f(0) = 0$, $f(1) = 1$: $M = 5$, $m = 0$, so $M + m = 5$.
13	**(b)**	$f' = \sin x(2\cos x - 1) = 0$: $c = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$, sum 3π.
14	**(e)**	$a = \frac{\pi}{6}$, $f'(a) = \sec^2\frac{\pi}{6} = \frac{4}{3}$, so $\left(f^{-1}\right)' = \frac{3}{4}$.